

EXERCISE 7

Type 1.

1. Prove that $\frac{2\sin\theta - \sin 2\theta}{2\sin\theta + \sin 2\theta} = \tan^2 \frac{\theta}{2}$
2. Prove that $\cot \frac{\theta}{2} - \tan \frac{\theta}{2} = 2\cot\theta$
3. Prove that $\frac{1 + \sin\theta}{1 - \sin\theta} = \tan^2 \left(\frac{\pi}{4} + \frac{\theta}{2} \right)$
4. Prove that $\sec\theta + \tan\theta = \tan \left(\frac{\pi}{4} + \frac{\theta}{2} \right)$ [S. S. 54, 57]
5. Prove that $\frac{\sin\alpha + \sin\beta - \sin(\alpha + \beta)}{\sin\alpha + \sin\beta + \sin(\alpha + \beta)} = \tan \frac{\alpha}{2} \tan \frac{\beta}{2}$ [P. U. 57]
6. Prove that $\tan \left(\frac{\pi}{4} - \frac{A}{2} \right) = \sec A - \tan A = \sqrt{\frac{1 - \sin A}{1 + \sin A}}$
7. Prove that $\operatorname{cosec} \left(\frac{\pi}{4} + \frac{\theta}{2} \right) \operatorname{cosec} \left(\frac{\pi}{4} - \frac{\theta}{2} \right) = 2\sec\theta$
8. Prove that $(\cos A - \cos B)^2 + (\sin A - \sin B)^2 = 4\sin^2 \frac{A - B}{2}$
9. Prove that $\cos^2 \frac{\pi}{8} + \cos^2 \frac{3\pi}{8} + \cos^2 \frac{5\pi}{8} + \cos^2 \frac{7\pi}{8} = 2$
10. Prove that $\sin^4 \frac{\pi}{8} + \sin^4 \frac{3\pi}{8} + \sin^4 \frac{5\pi}{8} + \sin^4 \frac{7\pi}{8} = \frac{3}{2}$ [Roorkee 80]
11. Prove that $\left(1 + \cos \frac{\pi}{8} \right) \left(1 + \cos \frac{3\pi}{8} \right) \left(1 + \cos \frac{5\pi}{8} \right) \left(1 + \cos \frac{7\pi}{8} \right)$
 $= \frac{1}{8}$ [I. I. T. 84]

Type II.

12. Show that $\cot 142\frac{1}{2}^\circ = \sqrt{2} + \sqrt{3} - 2 - \sqrt{6}$

[Hint : $\cot 142\frac{1}{2}^\circ = \cot(180^\circ - 37\frac{1}{2}^\circ) = -\cot 37\frac{1}{2}^\circ$]

13. Prove that $\sin^2 48^\circ - \cos^2 12^\circ = \frac{\sqrt{5} + 1}{8}$

14. Prove that $4(\sin 24^\circ + \cos 6^\circ) = \sqrt{3} + \sqrt{15}$

[Hint : L.H.S. = $4(\sin 24^\circ + \sin 84^\circ)$]

15. Prove that $\cot 6^\circ \cot 42^\circ \cot 66^\circ \cot 78^\circ = 1$ [P. U. 67; H, S, 66, 70]

16. Prove that $\tan 12^\circ \tan 24^\circ \tan 48^\circ \tan 84^\circ = 1$

17. Prove that $\sin 6^\circ \sin 42^\circ \sin 66^\circ \sin 78^\circ = \frac{1}{16}$

18. Prove that $\sin \frac{\pi}{5} \sin \frac{2\pi}{5} \sin \frac{3\pi}{5} \sin \frac{4\pi}{5} = \frac{5}{16}$

19. Prove that $\cos 36^\circ \cos 72^\circ \cos 108^\circ \cos 144^\circ = \frac{1}{16}$

20. Prove that $\cos \frac{\pi}{15} \cos \frac{2\pi}{15} \cos \frac{3\pi}{15} \cos \frac{4\pi}{15} \cos \frac{5\pi}{15} \cos \frac{6\pi}{15} \cos \frac{7\pi}{15} = \frac{1}{2^7}$

21. Show that $\cos \frac{\pi}{65} \cos \frac{2\pi}{65} \cos \frac{4\pi}{65} \cos \frac{8\pi}{65} \cos \frac{16\pi}{65} \cos \frac{32\pi}{65} = \frac{1}{64}$ [M. N. R. '97]

Type III.

22. If $\tan \frac{\theta}{2} = \sqrt{\frac{1-e}{1+e}} \tan \frac{\varphi}{2}$, prove that $\cos \varphi = \frac{\cos \theta - e}{1 - e \cos \theta}$ [P. U. 62]

23. If $\sin \alpha + \sin \beta = a$ and $\cos \alpha + \cos \beta = b$, prove that

(i) $\sin(\alpha + \beta) = \frac{2ab}{a^2 + b^2}$

[P. U. 50]

(ii) $\cos(\alpha - \beta) = \frac{1}{2}(a^2 + b^2 - 2)$

[P. U. 41]

24. If α and β be the two different roots of equation $a \cos \theta + b \sin \theta = c$, prove that

(i) $\tan(\alpha + \beta) = \frac{2ab}{a^2 - b^2}$

(ii) $\cos(\alpha + \beta) = \frac{a^2 - b^2}{a^2 + b^2}$

25. If $\cos \alpha + \cos \beta = \frac{1}{3}$ and $\sin \alpha + \sin \beta = \frac{1}{4}$, prove that

$\cos \frac{\alpha - \beta}{2} = \pm \frac{5}{24}$

[Inter 85]



26. If $2\tan \frac{\alpha}{2} = \tan \frac{\beta}{2}$, prove that $\cos \alpha = \frac{3 + 5\cos \beta}{5 + 3\cos \beta}$

27. If $\sin \alpha = \frac{4}{5}$ and $\cos \beta = \frac{5}{13}$, prove that $\cos \frac{\alpha - \beta}{2} = \frac{8}{\sqrt{65}}$

28. If $\sec(\varphi + \alpha) + \sec(\varphi - \alpha) = 2\sec \varphi$, prove that $\cos \varphi = \sqrt{2} \cos \frac{\alpha}{2}$

[P. U. 60, 71]

29. If $\cos \theta = \frac{\cos \alpha \cos \beta}{1 - \sin \alpha \sin \beta}$, prove that

$$\tan \frac{\theta}{2} = \frac{\tan \frac{\alpha}{2} - \tan \frac{\beta}{2}}{1 - \tan \frac{\alpha}{2} \tan \frac{\beta}{2}}$$

30. If $\tan \alpha = \frac{\sin \theta \sin \varphi}{\cos \theta + \cos \varphi}$, prove that one of the values of $\tan \frac{\alpha}{2}$ is

$$\tan \frac{\theta}{2} \tan \frac{\varphi}{2}.$$

31. If $\cos \theta = \frac{\cos \alpha + \cos \beta}{1 + \cos \alpha \cos \beta}$, prove that one of the values of $\tan \frac{\theta}{2}$ is

$$\tan \frac{\alpha}{2} \tan \frac{\beta}{2}.$$

